CS471 – Lesson 13 – Reinforcement Learning
Value Iteration (Instructor Solution Key)

In micro-blackjack, you repeatedly draw a card (with replacement) that is equally likely to be a 2, 3, or 4. You can either Draw or Stop if the total score of the cards you have drawn is less than 6. If your total score is 6 or higher, the game ends, and you receive a utility of 0. When you Stop, your utility is equal to your total score (up to 5), and the game ends. When you Draw, you receive no utility. There is a discount factor ($\gamma = 0.9$). Let's formulate this problem as an MDP with the following states: 0, 2, 3, 4, 5, and a Done state, for when the game ends.

1. Fill in the following table of value iteration values for the first 5 iterations, or if it converges only complete the table to convergence.

States	**0**	**2**	**3**	**4**	**5**
V_0	0	0	0	0	0
V_1	0	2	3	4	5
V_2	2.7	2.7	3	4	5
V_3	2.91	2.7	3	4	5
V_4	2.91	2.7	3	4	5
V_5	2.91	2.7	3	4	5

Note: The values completely converge at V_3.

Step by Step - General Bellman Update Equation:

$$\text{For any state } s \in \{0, 2, 3, 4, 5\}:$$

$$V_k(s) = \max(\text{Stop}(s), \text{Draw}(s))$$

$$\text{Where: } \text{Stop}(s) = s, \quad \text{Draw}(s) = 0.9 \sum_{c \in \{2,3,4\}} \frac{1}{3} V_{k-1}(s+c)$$

Note: If $s + c \geq 6$, the game transitions to the Done state, where $V(s') = 0$.

Initial State ($k = 0$)

$$V_0(0) = 0, \quad V_0(2) = 0, \quad V_0(3) = 0, \quad V_0(4) = 0, \quad V_0(5) = 0$$

Iteration 1 ($k = 1$)

Since $V_0(s) = 0$ for all states, $\text{Draw}(s) = 0$ for all s.

- $V_1(0) = \max(0, 0) = 0$
- $V_1(2) = \max(2, 0) = 2$
- $V_1(3) = \max(3, 0) = 3$

- $V_1(4) = \max(4, 0) = 4$
- $V_1(5) = \max(5, 0) = 5$

Iteration 2 ($k = 2$)

- **State 0:**

$$\text{Draw}(0) = 0.9 \cdot \frac{V_1(2) + V_1(3) + V_1(4)}{3} = 0.9 \cdot \frac{2+3+4}{3} = 2.7$$
$$V_2(0) = \max(0, 2.7) = 2.7$$

- **State 2:**

$$\text{Draw}(2) = 0.9 \cdot \frac{V_1(4) + V_1(5) + V_1(\geq 6)}{3} = 0.9 \cdot \frac{4+5+0}{3} = 2.7$$
$$V_2(2) = \max(2, 2.7) = 2.7$$

- **State 3:**

$$\text{Draw}(3) = 0.9 \cdot \frac{V_1(5) + V_1(\geq 6) + V_1(\geq 6)}{3} = 0.9 \cdot \frac{5+0+0}{3} = 1.5$$
$$V_3(3) = \max(3, 1.5) = 3$$

- **States 4 & 5:** Drawing yields 0 expected utility because all outcome states are ≥ 6.

$$V_2(4) = \max(4, 0) = 4, \quad V_2(5) = \max(5, 0) = 5$$

Iteration 3 ($k = 3$)

- **State 0:** Recalculated due to updated value $V_2(2) = 2.7$:

$$\text{Draw}(0) = 0.9 \cdot \frac{V_2(2) + V_2(3) + V_2(4)}{3} = 0.9 \cdot \frac{2.7+3+4}{3} = 0.9 \cdot \frac{9.7}{3} = 2.91$$
$$V_3(0) = \max(0, 2.91) = 2.91$$

- **States 2, 3, 4, 5:** All values remain unchanged from V_2:

$$V_3(2) = 2.7, \quad V_3(3) = 3, \quad V_3(4) = 4, \quad V_3(5) = 5$$

Iteration 4 ($k = 4$ − Verification of Convergence)

- **State 0:**

$$\text{Draw}(0) = 0.9 \cdot \frac{V_3(2) + V_3(3) + V_3(4)}{3} = 0.9 \cdot \frac{2.7+3+4}{3} = 2.91$$
$$V_4(0) = V_3(0) = 2.91$$

Since $V_4(s) = V_3(s)$ for all states s, the algorithm has fully converged at $k = 3$.

2. What is the optimal policy for this MDP?

States	**0**	**2**	**3**	**4**	**5**
π^*	Draw	Draw	Stop	Stop	Stop